KEITH FRANCIS

Software Engineer | U.S. Marine Corps Veteran

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SUMMARY

Software engineer and 12-year Marine Corps veteran with 3.5 years building and hardening an internal platform at Capital One. Focused on reliability: framework migrations, test automation, and resiliency testing, including taking a platform's failover pass rate from 50% to 100%. Builds full-stack TypeScript applications, using AI tools like Claude Code with a tester's discipline. Background in avionics test equipment calibration and Lean Six Sigma instruction; previously held a Top Secret clearance.

WORK EXPERIENCE

Full-Stack Engineer — Capital One (Contractor through Cognixia) Mar 2022 – Sep 2025

McLean, VA (Remote)

- Supported "Foundation," an internal React/Vue platform hosting partner apps that teams across Capital One used to coordinate customer sales and communication events.
- Led the Node.js 14 → 18 migration and guided teams through Node 18 → 20; helped move the build to Vite, cutting deployments by ~15 minutes.
- Raised unit test coverage (Jest, Vitest) from 80% to 95%, with 100% end-to-end coverage in Cypress.
- Owned regional isolation (failover) testing, a regulatory requirement for banks: pass rate 50% → 100%, isolation rating 15-minute (Gold) → 5-minute (Platinum).
- Co-developed a Python script automating isolation checks in AWS and Cloud Doctor, cutting each run from ~15 minutes to ~2.
- Fixed dozens of bugs and security vulnerabilities across Foundation and sister-team apps; mentored 4 associates, 2 promoted to Senior Software Engineer.

Sergeant (E-5) — United States Marine Corps Jun 2002 – Dec 2014

Jacksonville, NC

Calibration Technician (8 years)

- Calibrated and repaired avionics test equipment for fixed- and rotary-wing aircraft, from gauge blocks and torque wrenches to frequency generators, engine test sets, air data, radio, radar and weapons systems.
- Level 4 qualified in Physical/Dimensional, Torque/Pressure, Electrical/Electronic and Aircraft Engine Test Systems; Level 2 in Microwave.

Lean Six Sigma Senior Instructor (4 years)

- Senior instructor for 2.5 years in the Navy and Marine Corps Continuous Process Improvement program, training new instructors and Yellow and Green Belt practitioners.
- Authored training programs, built process-evaluation tools to target projects, and created a just-in-time refresher system.

SELECTED PROJECTS

HackerSocket — Developer Social Platform and Tiered Job Board github.com/v8integra/hackersocket

Next.js, TypeScript, PostgreSQL, Prisma

- Built from scratch: credentials and GitHub OAuth with account linking, a PostgreSQL/Prisma data layer, a feed, follows, direct messaging, and a job board split into entry, mid and senior tiers.
- Added self-service account deletion and verified-email linking to meet state privacy laws (CCPA and similar).

DocStew — Local-First AI Document Studio github.com/v8integra/DocStew

TypeScript, Electron, SQLite, Ollama

- Plugin-based editor for 9 document types sharing one local-AI layer (Ollama) for retrieval-augmented Q&A, summarization and translation; 284 passing automated tests.

- Built with Claude Code and verified against the running app, catching real defects (CSP violations, native-module ABI mismatches, race conditions) before shipping a Windows installer.

EDUCATION

M.S. Information and Communications Technology – Web Design and Development 2021
University of Denver

B.S. Computer Science – Human-Centered Computing 2019
Colorado State University

CERTIFICATIONS AND CLEARANCE

- **Lean Six Sigma Green Belt** – Department of Defense, 2014 (completed Black Belt-level training)
- **AI Engineering, AI for Web Developers** – Scrimba, 2026 · **Google AI** – Google, 2026
- **Security clearance:** previously held Secret and Top Secret (inactive since 2015)

TECHNICAL SKILLS

- **Languages and frameworks:** TypeScript, JavaScript, Python, Java, SQL, React, Vue.js, Next.js, Node.js, Express.js, Electron
- **Testing and build:** Jest, Vitest, Cypress, Vite, Git, GitHub
- **Cloud, data and AI:** AWS, PostgreSQL, Prisma, SQLite, Claude Code, Ollama (local LLMs), RAG